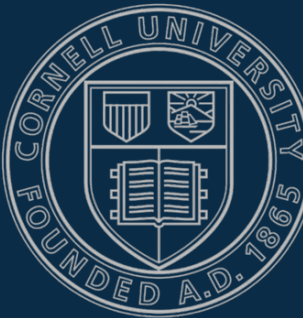


Kalman Down the Noise: Teaching Bayes What to Trust

Reimplementing *Weight Uncertainty in Neural Networks* (Blundell et al., 2015) on MNIST with a Kalman gradient-trust extension.



01 — MOTIVATION

Goal: Learning Weight Distributions

Standard neural networks learn fixed weights, while **Bayes by Backprop (BBB)** learns a distribution over each weight. By sampling weights during each forward pass, BBB produces uncertainty-aware predictions.

02 — METHODOLOGY

Pulling Weights from a Distribution

In a normal neural network, each weight is a single learned number. In Bayes by Backprop, each weight is treated as a learned distribution (usually a diagonal Gaussian):

$$w \sim q(w|\theta), q(w|\theta) = \mathcal{N}(w|\mu, \sigma^2)$$

Each weight stores a mean μ and an uncertainty parameter ρ . The standard deviation is defined as:

$$\sigma = \ln(1 + e^\rho) \Rightarrow \sigma \geq 0 \quad \forall \quad \rho$$

To make this trainable, the paper uses the reparameterization trick:

$$w \sim \mathcal{N}(w|\mu, \sigma^2) \equiv w \sim \mu + \sigma \varepsilon \text{ where, } \varepsilon \sim \mathcal{N}(0, 1)$$

Loss Function—Cross Entropy Loss:

$$F(D, \theta) = KL[q(w|\theta)||P(w)] - \langle \ln(P(D|w)) \rangle_{q(w|\theta)}$$

- **KL: complexity cost**
 - Penalizes weight distributions that drift too far from the prior.
- **Likelihood cost**
 - Typical data-fitting part: $\mathcal{L}_{BBB} \sim$ complexity cost + data fitting cost

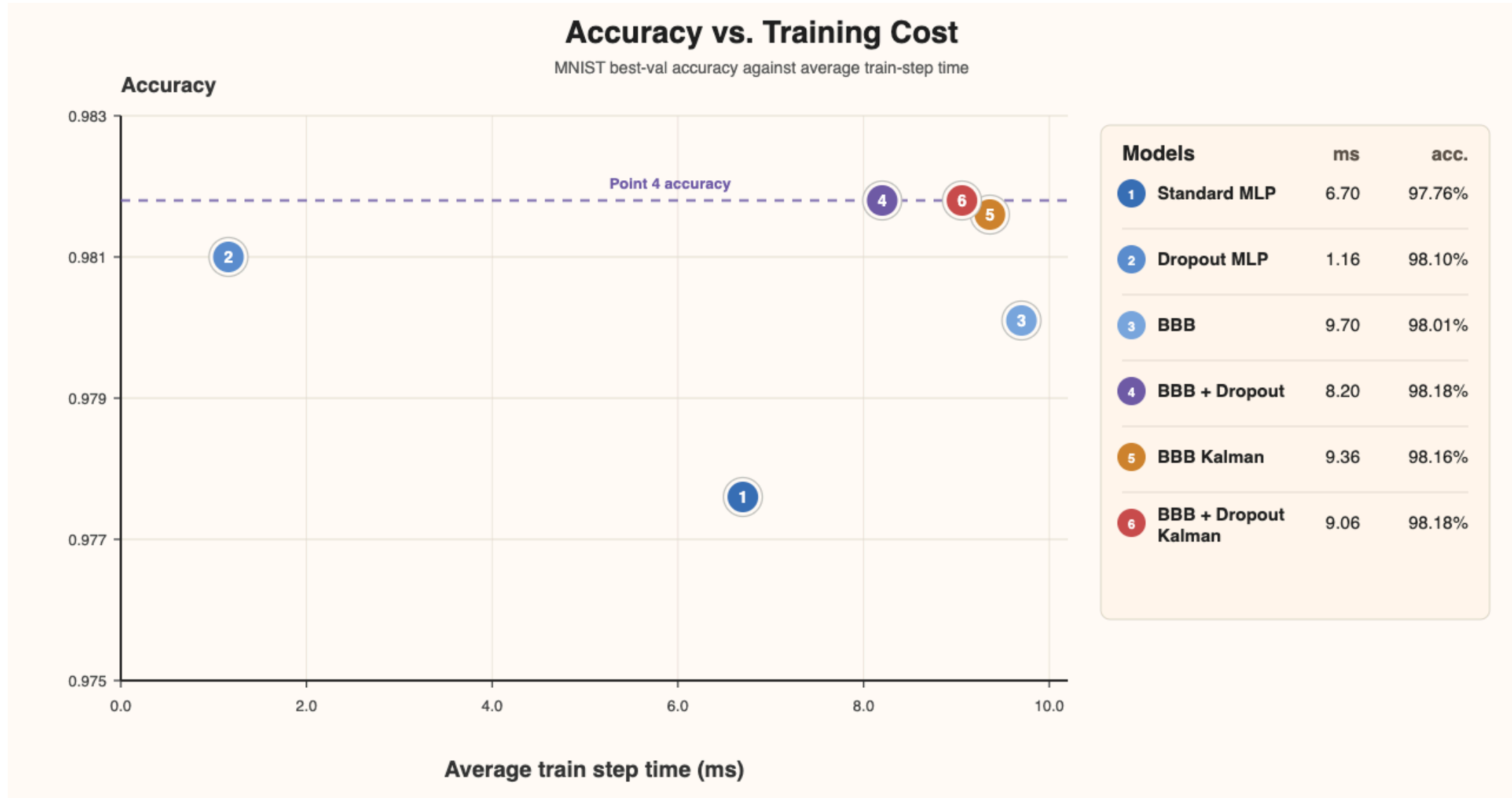


FIGURE 2 · ACCURACY VS. TRAINING COST
MNIST accuracy is compared against average training-step time for the baseline, BBB, dropout, and Kalman-extension models. Kalman stays competitive in accuracy but adds substantial training cost and roughly doubles memory use compared with dropout.

03 — RESULTS · MNIST

Reproducing the paper's training behavior

We match the paper’s model performance ordering despite a smaller architecture, suggesting we capture the same training dynamics.

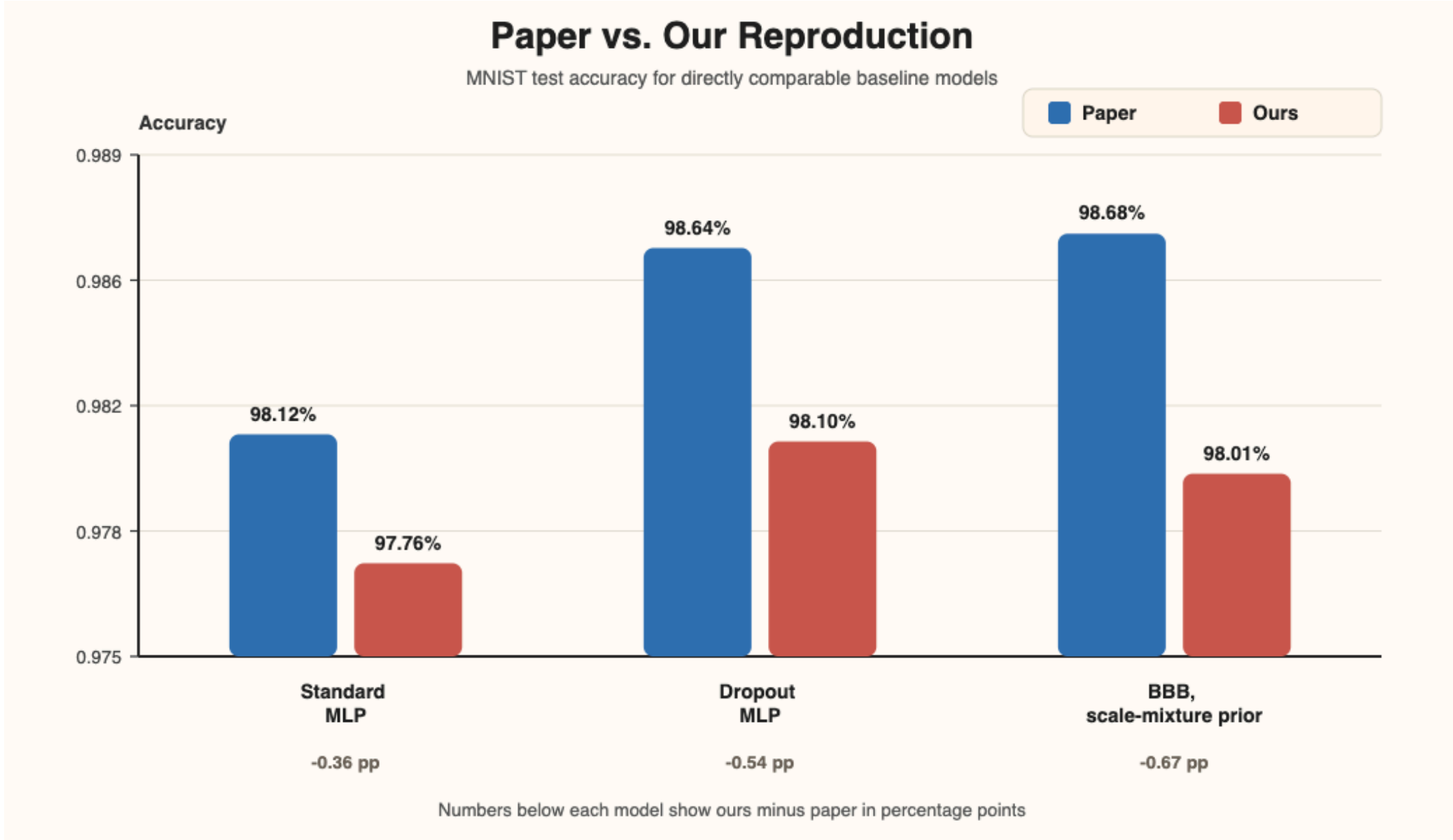


FIGURE 1 · PAPER VS. OUR REPRODUCTION
Our smaller h=400, L=2 MNIST models reproduce the relative performance trend from the Bayes by Backprop paper, using model ordering rather than exact accuracy values from the original 784 → 1200 → 1200 → 10 ReLU MLP benchmark.

04 — EXTENSIONS

Kalman Hypothesis — Kizzfinder Algorithm

Our Kalman trust step adds an estimate of uncertainty during optimization. Unstable gradients are dampend before Adam applies it.

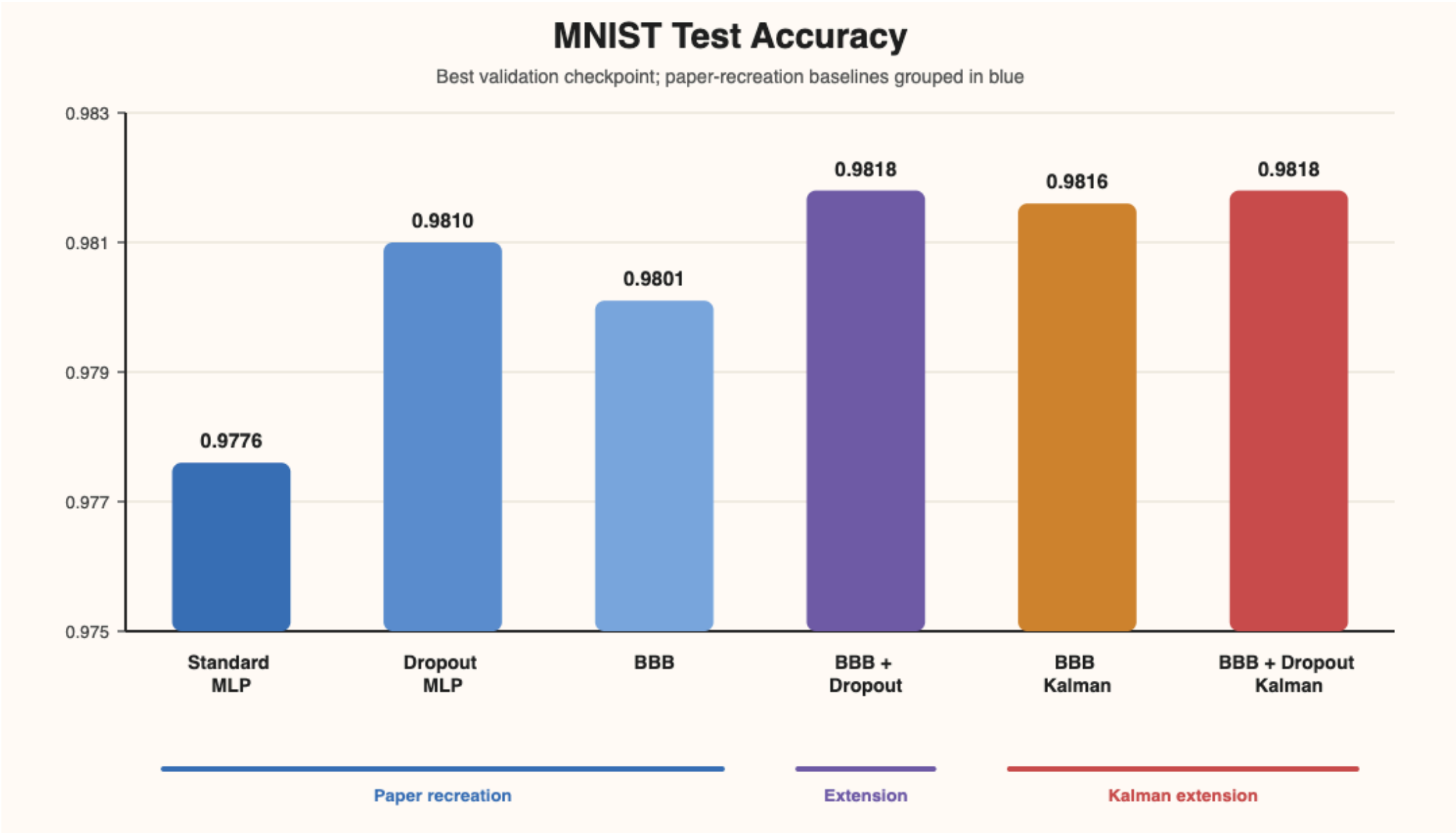


FIGURE 3 · MNIST TEST ACCURACY
MNIST test accuracy across paper-recreation baselines, BBB + Dropout, and Kalman extensions. BBB + Dropout and BBB + Dropout Kalman achieve the highest accuracy, while all models remain closely clustered.

05 — RESULTS

MNIST test accuracy

We trained each model across multiple seeds and compared mean test accuracy ±1 SD on clean MNIST and noisy FashionMNIST to evaluate whether Kalman-style updates improved over baseline models.

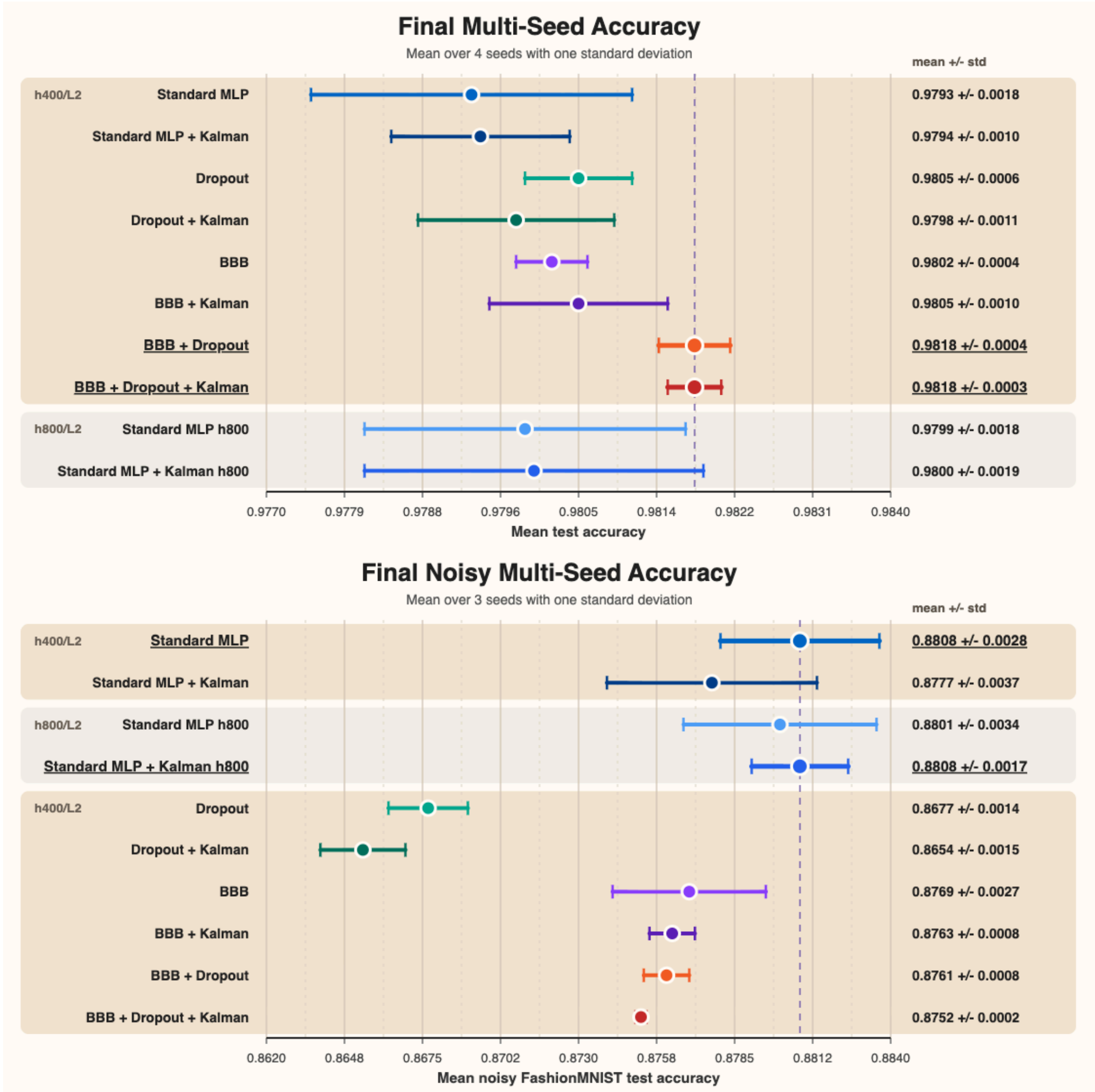


FIGURE 4 · FINAL MULTI-SEED ACCURACY
Mean multi-seed accuracy is compared across clean and noisy settings, showing the Kalman-augmented standard MLP as the strongest overall model and suggesting its main benefit is robustness.

07 — CONCLUSION

Summary

- Reproduced key trends from Bayes by Backprop
- Introduced a Kalman-style trust update as an extension
- Kalman-augmented MLP performed well across clean + noisy data
- Gains were modest relative to added computational cost

FUTURE WORK

Test Kalman trust on larger and deeper models, extend it to CNNs, and use it as a layerwise learning-rate controller rather than raw gradient scaling. Future variants should add direction-aware and per-neuron/filter trust, then compare against earlier Kalman-style methods. Further look at BBB with heteroscedastic weights.